

# Ian Q. Snider

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## EDUCATION

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**University of California, Berkeley, Berkeley, CA**

*Aug 2025 - present*

Graduate Researcher in Nuclear Engineering

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**Washington University in St. Louis, St. Louis, MO**

*May 2025*

B.S. Mechanical Engineering

4.00/4.00

**Truman State University, Kirksville, MO**

*Conferred: Dec 2024*

B.A. Physics, Mathematics Minor

3.89/4.00

## EXPERIENCE

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**Lawrence Berkeley National Laboratory - Lab Affiliate, Berkeley, CA**

*Aug 2025 - present*

- Graduate researcher in the Bay Area Neutron Group
- Researching experimental techniques for indirect neutron capture cross-section measurement of short-lived nuclei

**Lawrence Livermore National Laboratory - Graduate Intern, Livermore, CA**

*June 2025 - present*

- Developed CRISP (CRITICAL Simulation Pipeline) for parallelizing jobs from the radiation transport code: COG
- Test driven Python development included Poetry, CI/CD, SQLite, MPI, CLI, feature branching, and Sphinx
- Used 2 world-class LLNL HPC clusters Dane and Ruby with Slurm sbatch/srun scripts
- CRISP successfully automated the materials compiling and  $k_{\text{eff}}$  benchmarking of ~3420 ICSBEP critical reactors

**Brookhaven National Laboratory - Nuclear Science Intern, Upton, NY**

*2022 - 2024*

- Developed BRR (Bayesian Resonance Reclassifier) for heavy-nuclei resonance reclassification on NNDC clusters
- Applied Python machine learning using Scikit-learn and Pytorch to classify resonance spin assignments for capture cross-sections of In-115, Pb-206, and U-238
- Used nuclear data and Random Matrix Theory to develop synthetic training sets
- Used the neutron transport code OpenMC to perform perturbative sensitivity analyses of the thermal  $1/v$  neutron capture cross-sections for critical reactors on Gd-155, 157 and U-235, 238

**Truman State University - Student Astronomy Researcher, Kirksville, MO**

*2021 - 2022*

- Calculated trajectories for ~3000 Starlink satellites to optimize telescope viewing plans and developed a GUI
- Researched long-exposure luminosity data corruption due to satellite interference

## SKILLS

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- **Programming:** Python, Go, C, C++, Bash, Lua, SQL, MATLAB, Mathematica, Octave, LaTeX, Typst, Vimscript
- **Software/Technical:** Simulink, SolidWorks, OpenMC, COG, Pytorch, Scikit-learn, Slurm, Git, Linux, Arduino
- **Physics/engineering:** Electrodynamics, Electronics, Classical Mechanics, Quantum Physics, Mathematical Physics, Vibrations, Nuclear Physics, Thermodynamics, Fluid Mechanics, Solid Mechanics, Heat Transfer, Acoustics, Materials Science, Thermal Systems, Turbojets, Ramjets, Autonomous Aerial Vehicle Control
- **Mathematics:** Linear Algebra, ODEs, Computing Structures, Control Systems, Machine Learning, and Optimizations

## ACTIVITIES

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**WashU Robotics MATE ROV & MARINER Projects - Mechanical Lead**

*2023 - 2025*

- Collaborated with other students to develop an advanced autonomous underwater vehicle (AUV)
- Prototyped hydrodynamic dive control using an IMU and sonar state estimation with a Kalman Filter
- Built a chassis and buoyancy engine

**Society of Physics Students - Demo Chair**

*2020 - 2023*

- Developed 3 new physics demos, performed demos at meetings, and provided volunteer physics tutoring
- Wrote and proctored exams for 2022 & 2023 Science Olympiads ("Crave the Wave" and "Remote Sensing")